AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 30 · STUDENT LESSON

Agents and Tools

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Read the state. Check the edge. Respect the stop.

Trace bounded helpers with explicit graph rules, budgets and evidence limits.

Keep one habit: Declared graph → first edge check → budget → result validation → terminal or review.

MAIN LESSON · SECTION 1

Can this helper take the next step?

> Check the declared state, transition, tool contract and stopping rule.

Your learning goals

I can explain the **mechanism** and terminology of **state** machines for bounded **agent**s.

I can apply the **state**d procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall evidence and authority

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall **Evidence** and Sources: a retrieved record must support the precise claim. A tool result is **evidence** to validate, not permission for the next action.

Your end goal

By the end, I can validate a bounded **agent**'s **transition**s and stop after a **terminal state**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 30: **Agent**s and **Tool**s

An **agent** combines a goal, model, **tool**s, **state**, and a **stopping rule**.
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THINKER MISSION

Design a bounded library helper with three allowed **tool**s.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Name the stages and the allowed moves

> A bounded agent needs explicit limits, not just a goal.

The state-machine idea

A bounded **agent** uses explicit **state**s and finite action limits. A **state** machine makes invalid **transition**s visible.

Use constructed traces as evidence

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **agent**s and **tool**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Trace the declared graph before counting attempts

> A budget is a ceiling, not a new transition.

The original state and budget contract

States are ready, looking_up, review, done. ready may enter looking_up; looking_up may enter done or review; review may enter done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

Read the transition table

Current state	Allowed next state	Extra condition
ready	looking_up	approved read-only lookup; count an entry
looking_up	review or done	validate the returned result before choosing
review	done	authorised evidence required; no lookup retry edge
done	none	terminal within this run

Separate graph, budget and evidence checks

This declared graph has no retry cycle, so a legal run enters looking_up at most once. “Maximum two lookup attempts” is a separate upper limit, not permission to invent review→looking_up. A retry would require an explicitly changed graph and new tests. In this paper validator an attempt means entering looking_up; it is not a real external call. A transition-only check cannot verify retrieved facts or human authorisation without those inputs.

Make the deciding step visible

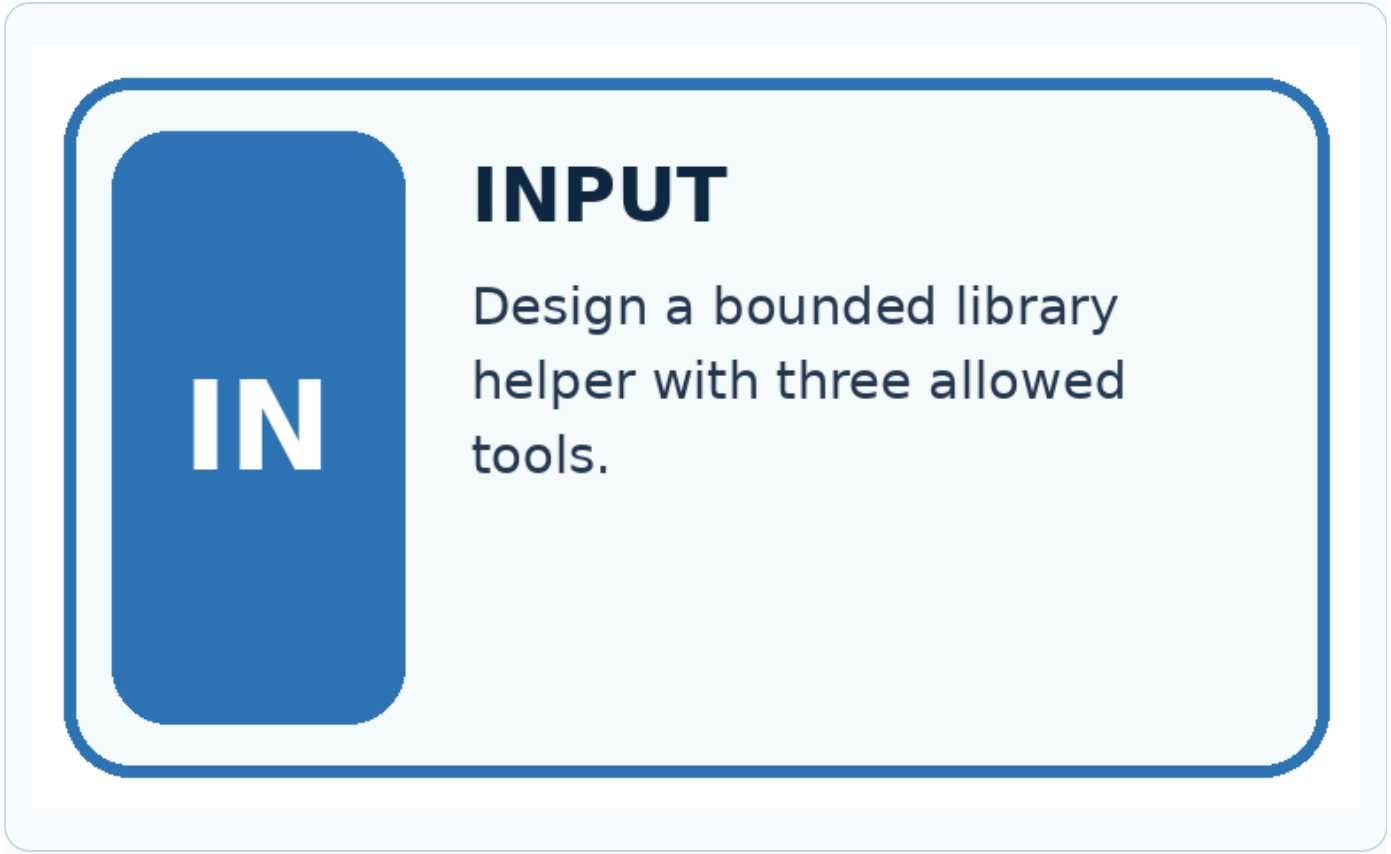
- Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → **Process** → **Output**

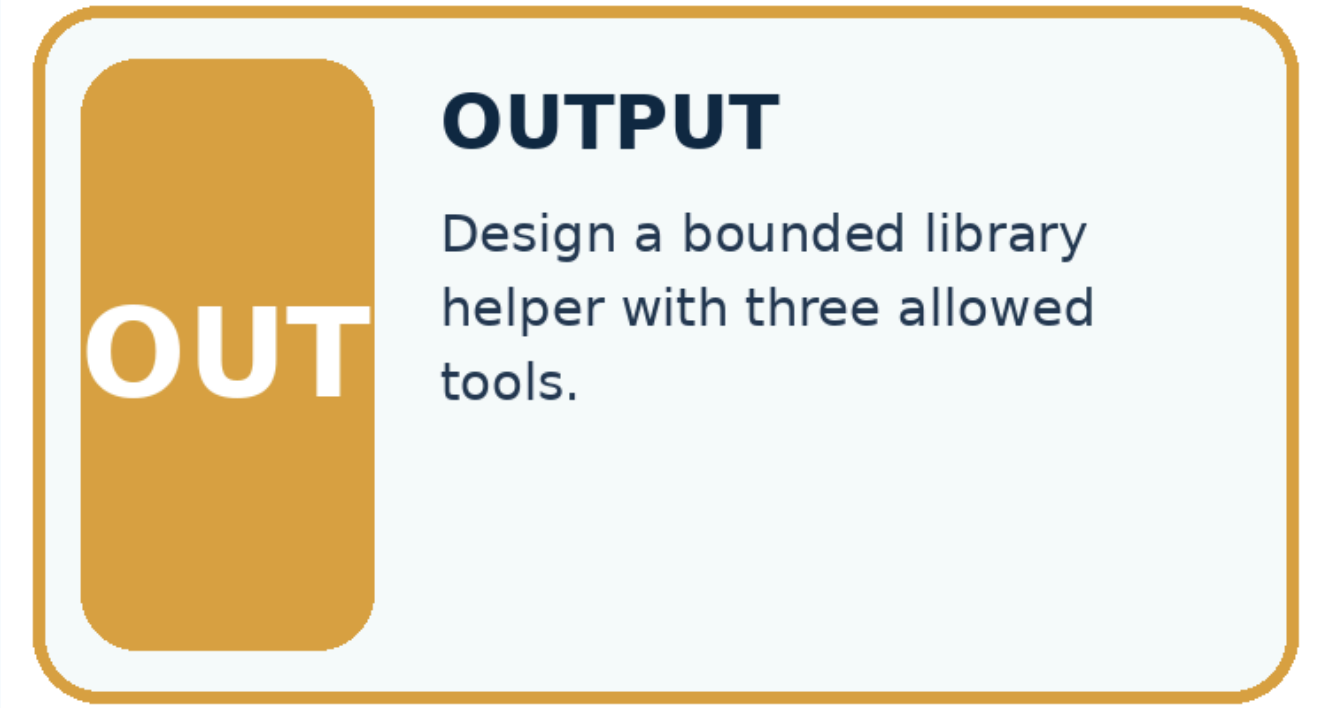
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Design a bounded library helper with three allowed tools.

↓


↻
PROCESS
An agent combines a goal, model, tools, state, and a stopping rule.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

State machines for bounded agents — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Trace ready -> looking_up -> review -> looking_up -> done.	States are ready, looking_up, review, done. ready may enter looking_up; looking_up may enter done or review; review may enter done after authorised evidence. done is terminal. Maximum two lookup attempts.	review -> looking_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the **state**d contract, record that condition explicitly instead of quietly replacing it with a convenient value.

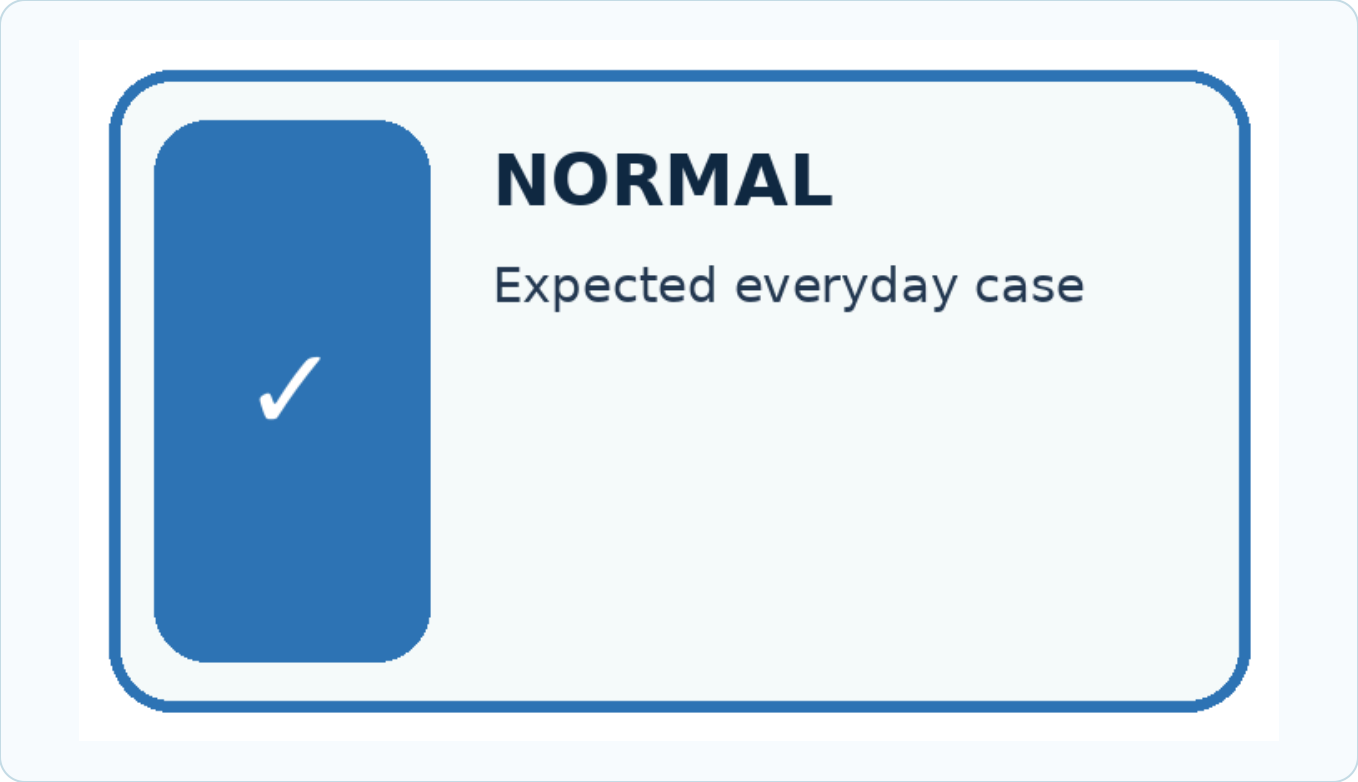
Connect the picture and the explanation

MAIN LESSON · SECTION 4

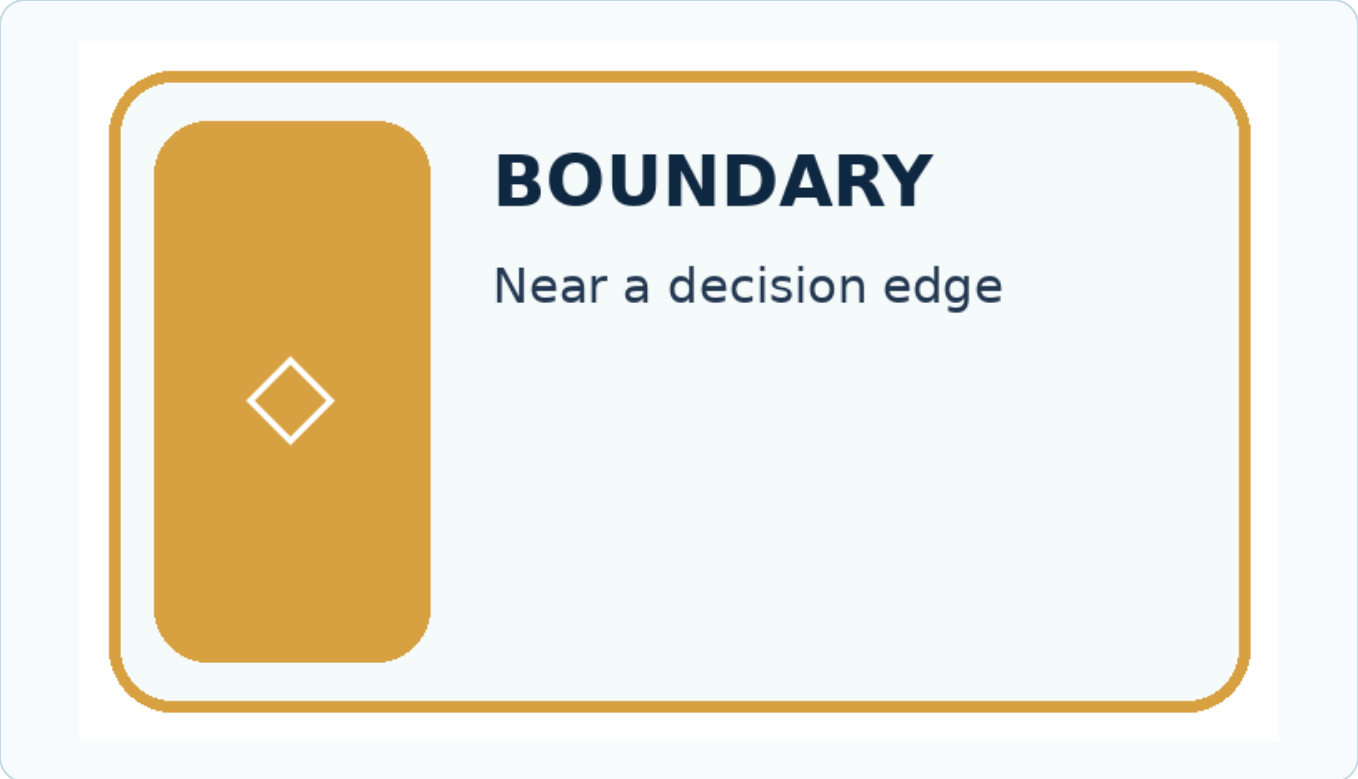
Compare valid steps, missing checks and stop conditions

> Finishing is insufficient if a required stage was skipped.

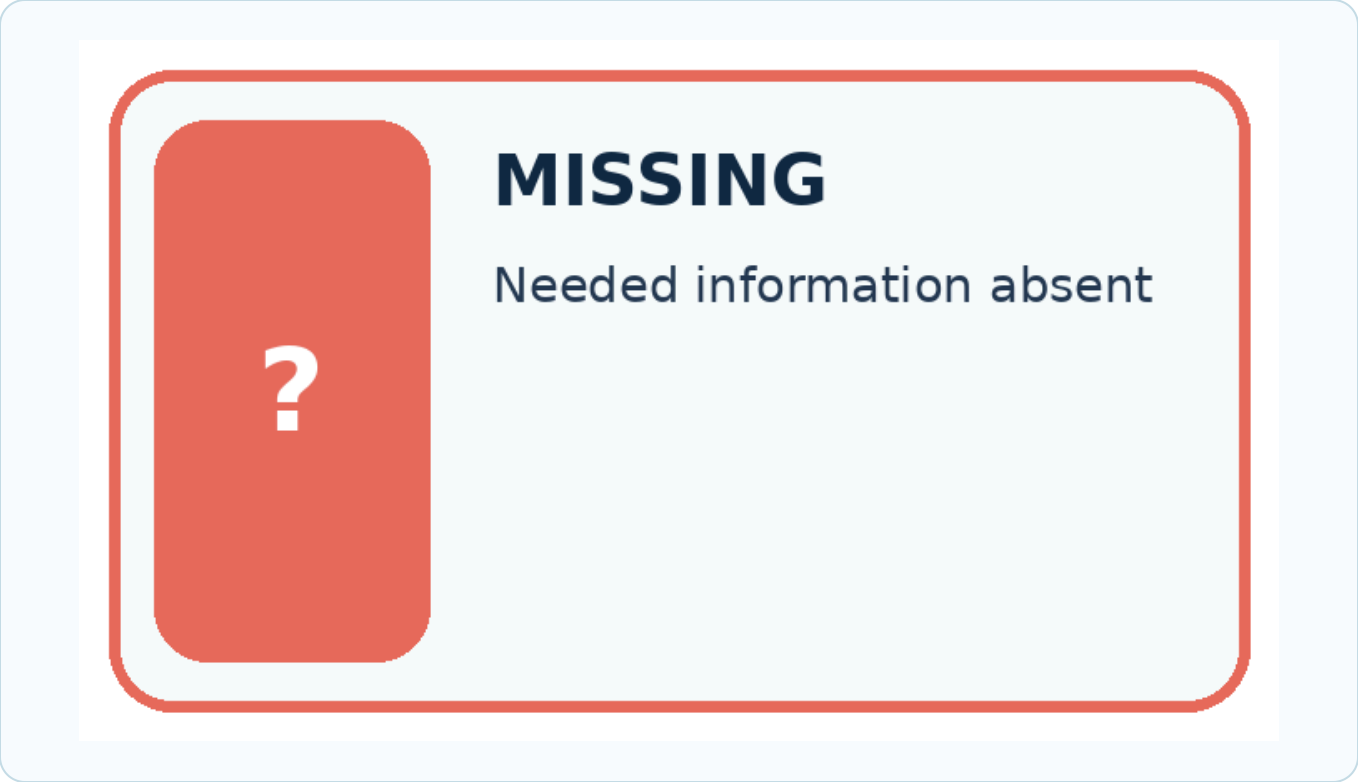
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Replace the edge checking→done with checking→review and review→done, without updating Trace A.

<details>
<summary>Check after predicting</summary>

Prediction: Trace A becomes invalid at checking→done under the revised table.

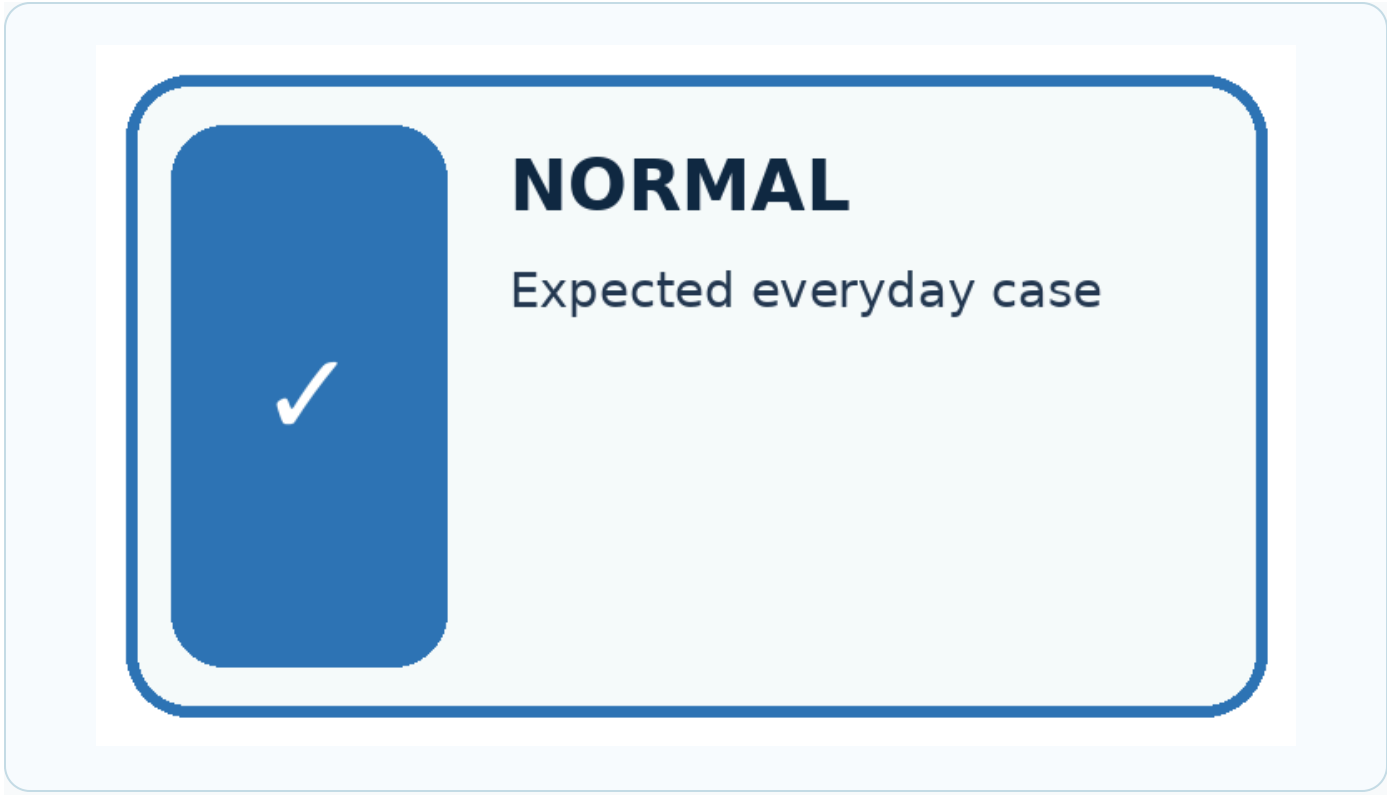
Reason: A required new **transition** changes what constitutes a complete run.
</details>

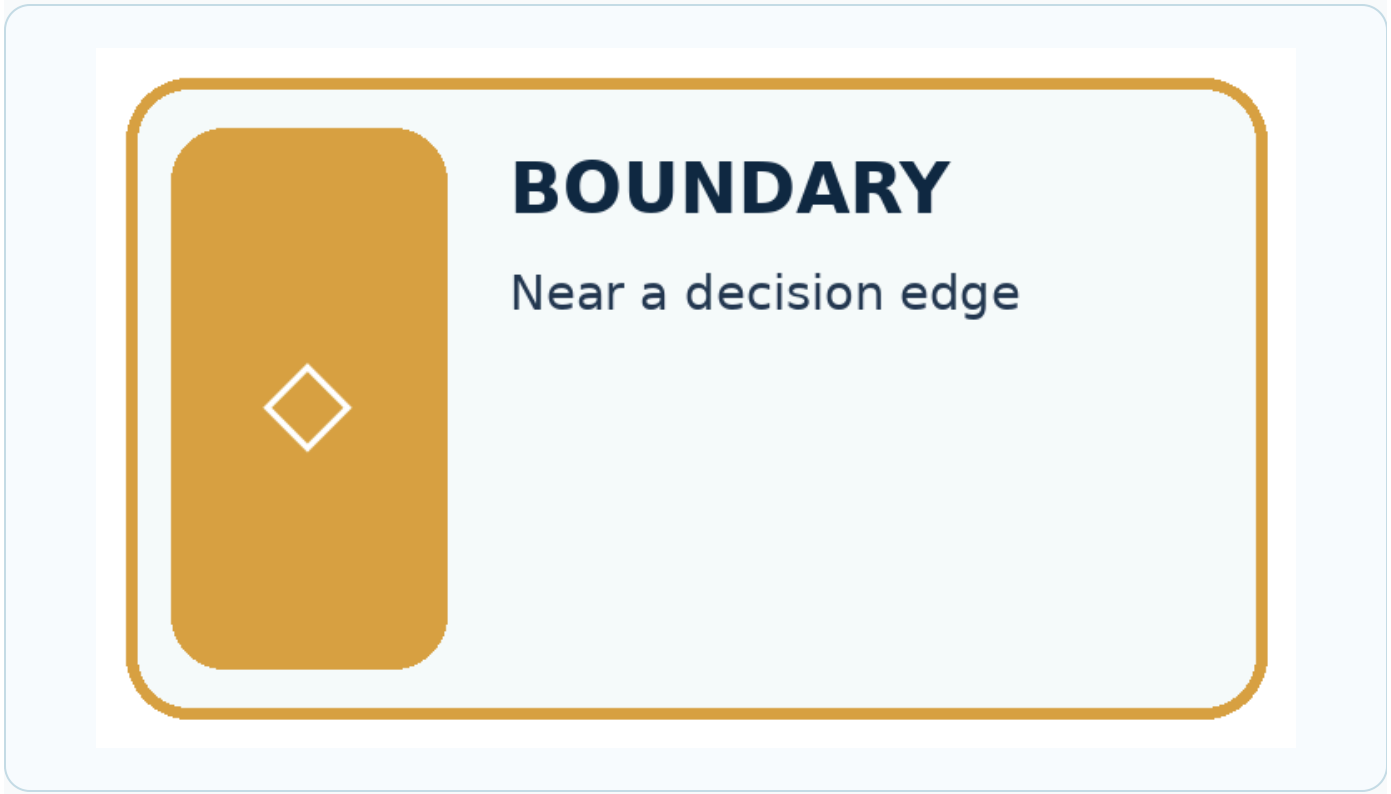
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<summary>Earlier wording, alternatives and picture transcripts</summary>

CASE GALLERY

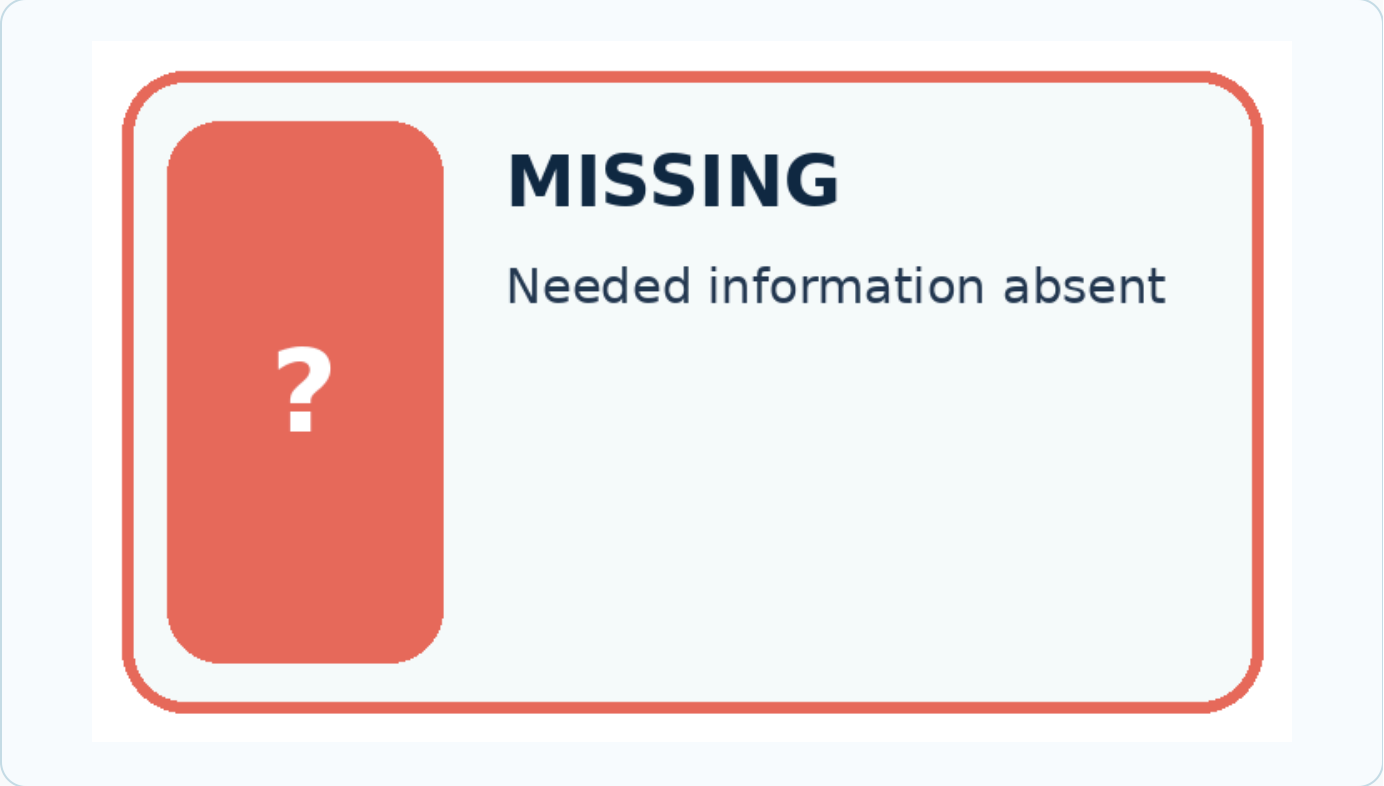
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.
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NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the first invalid edge, then test complete local code

> Keep the original ready/looking_up graph separate from the pictured searching/checking graph.

Try the original worked trace

- Trace ready -> looking_up -> review -> looking_up -> done.

▼ Check the first violation

review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Run the complete structural validator

```
ALLOWED = {'ready': {'looking_up'}, 'looking_up': {'review', 'done'},
           'review': {'done'}, 'done': set()}

def audit_trace(trace, budget=2):
    if type(budget) is not int or budget < 0:
        raise ValueError('budget must be a nonnegative integer')
    if not trace or trace[0] != 'ready':
        return 'invalid start'
    attempts = 0
    for before, after in zip(trace, trace[1:]):
        if after not in ALLOWED.get(before, set()):
            return f'invalid edge: {before}->{after}'
        if after == 'looking_up':
            if attempts >= budget:
                return 'lookup budget exceeded'
            attempts += 1
    return f"{'complete' if trace[-1] == 'done' else 'incomplete'}; lookups={attempts}"

for trace, budget in [(['ready','looking_up','done'], 2),
                      (['ready','looking_up','review','looking_up','done'], 2),
```

```
                    (['ready','looking_up','done'], 0),
                    (['ready','looking_up'], 2)]:
    print(audit_trace(trace, budget))
```

Interpret the checks and their limits

Predict the four printed lines: complete; lookups=1, invalid edge: review->looking_up, lookup budget exceeded, incomplete; lookups=1. The function checks graph edges and a lookup-entry budget only. It accepts review→done structurally but does not verify the required authorised evidence; no tool output, external call or authorisation record is simulated. “complete” means the structural trace ended at done, not that the goal is correct or an action is authorised.

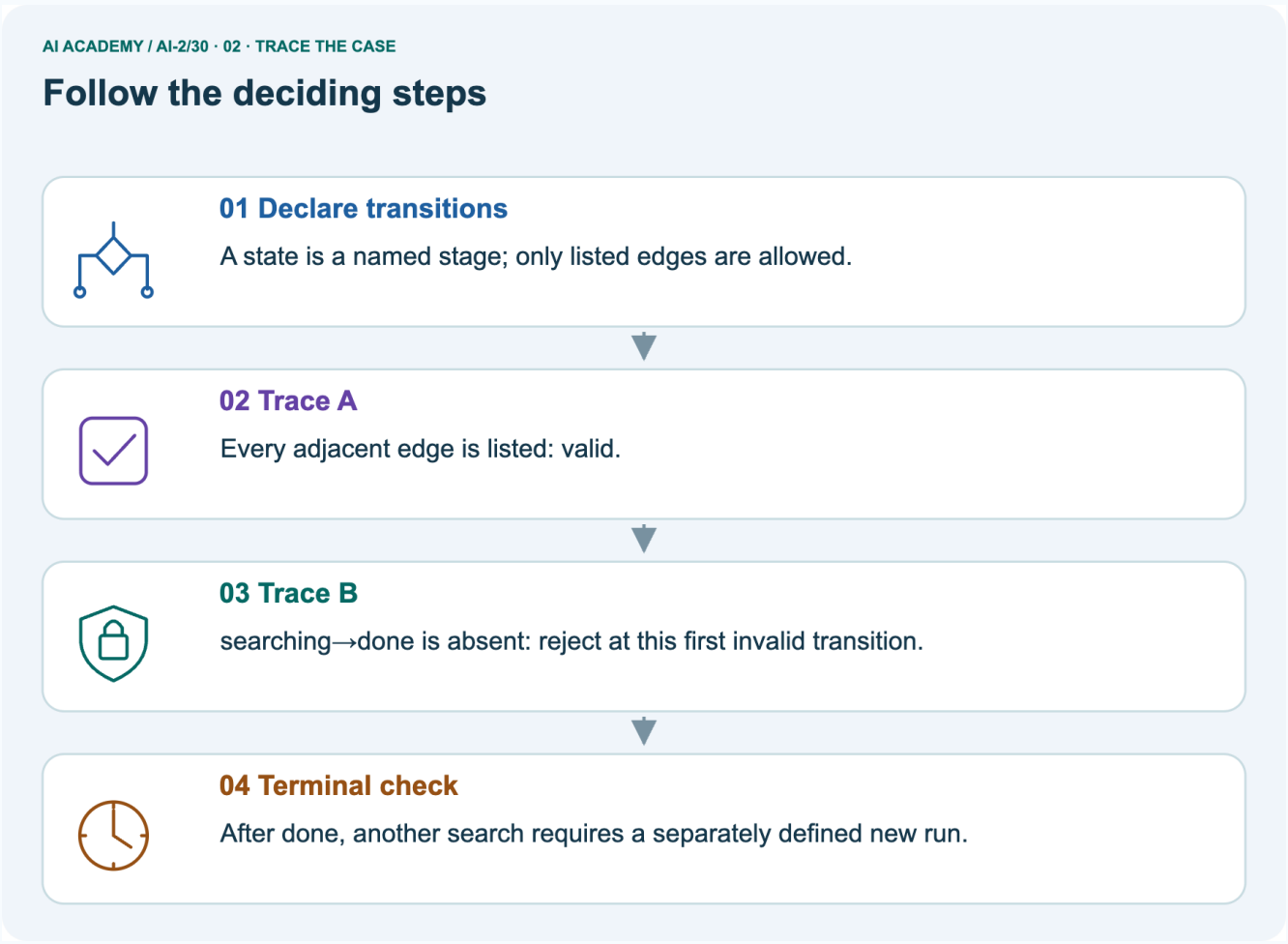
Read the separate pictured graph and traces

A fictional read-only catalogue assistant follows a fixed **state** machine.

Allowed edges: ready→searching; searching→checking; checking→done. done has no outgoing edge.

Trace A: ready→searching→checking→done. Trace B: ready→searching→done.

Trace the required checking stage



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare transitions	A state is a named stage; only listed edges are allowed.
2. Trace A	Every adjacent edge is listed: valid.
3. Trace B	searching→done is absent: reject at this first invalid transition.
4. Terminal check	After done, another search requires a separately defined new run.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

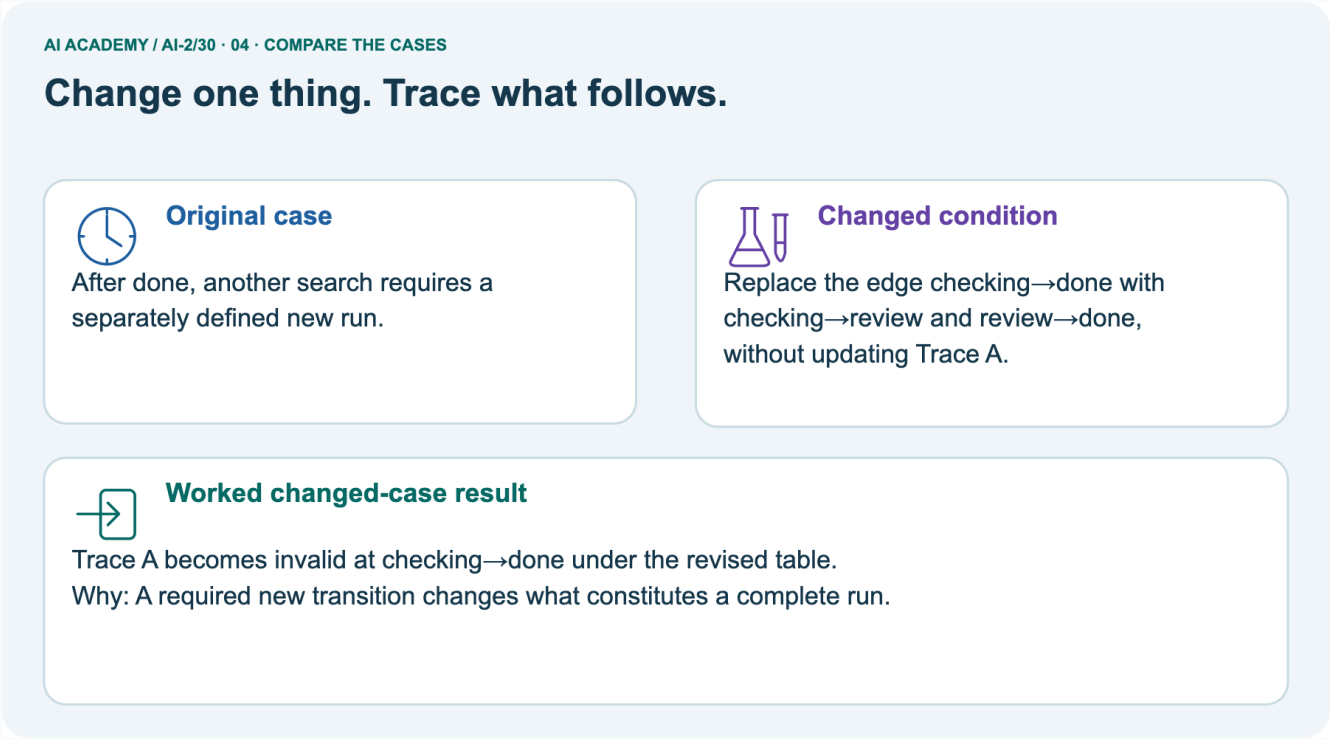
Reaching the desired ending is insufficient if the **agent** skipped a required check. Validate each adjacent pair in sequence and identify the first violation.

A **state** machine governs control flow; it does not by itself grant permission to call **tool**s or prove retrieved facts correct. Those are separate checks inside the relevant stages.

Predict a changed graph

- Change one thing: Replace the edge checking→done with checking→review and review→done, without updating Trace A. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the pictured limit

- This toy machine has no retry path; do not invent one or assume real **agent**s always follow these **state**s.
- Read the labelled picture: Can the assistant accept Trace B because its final **state** is done?

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 30 • WORKED EVIDENCE

1. Goal: help find a library book.
2. Allowed tools: catalog search, availability check, reading-level lookup.
3. State: request, candidates, evidence, attempts.
4. Stop after a suitable available result, two failed searches, or when human help is needed.

1. Goal: help find a library book.

2. Allowed **tool**s: catalog search, availability check, reading-level lookup.

3. **State**: request, candidates, **evidence**, attempts.

4. Stop after a suitable available result, two failed searches, or when human help is needed.

Recalculate, retrace, or re**state** the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final **state**ment. Explain why each number or decision follows from the **state**d **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and **state** the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def transition(state, target):
    allowed = {'ready': {'looking_up'}, 'looking_up': {'review', 'done'}, 'review':
{'done'}, 'done': set()}
    if target not in allowed.get(state, set()):
        raise ValueError('invalid transition')
    return target

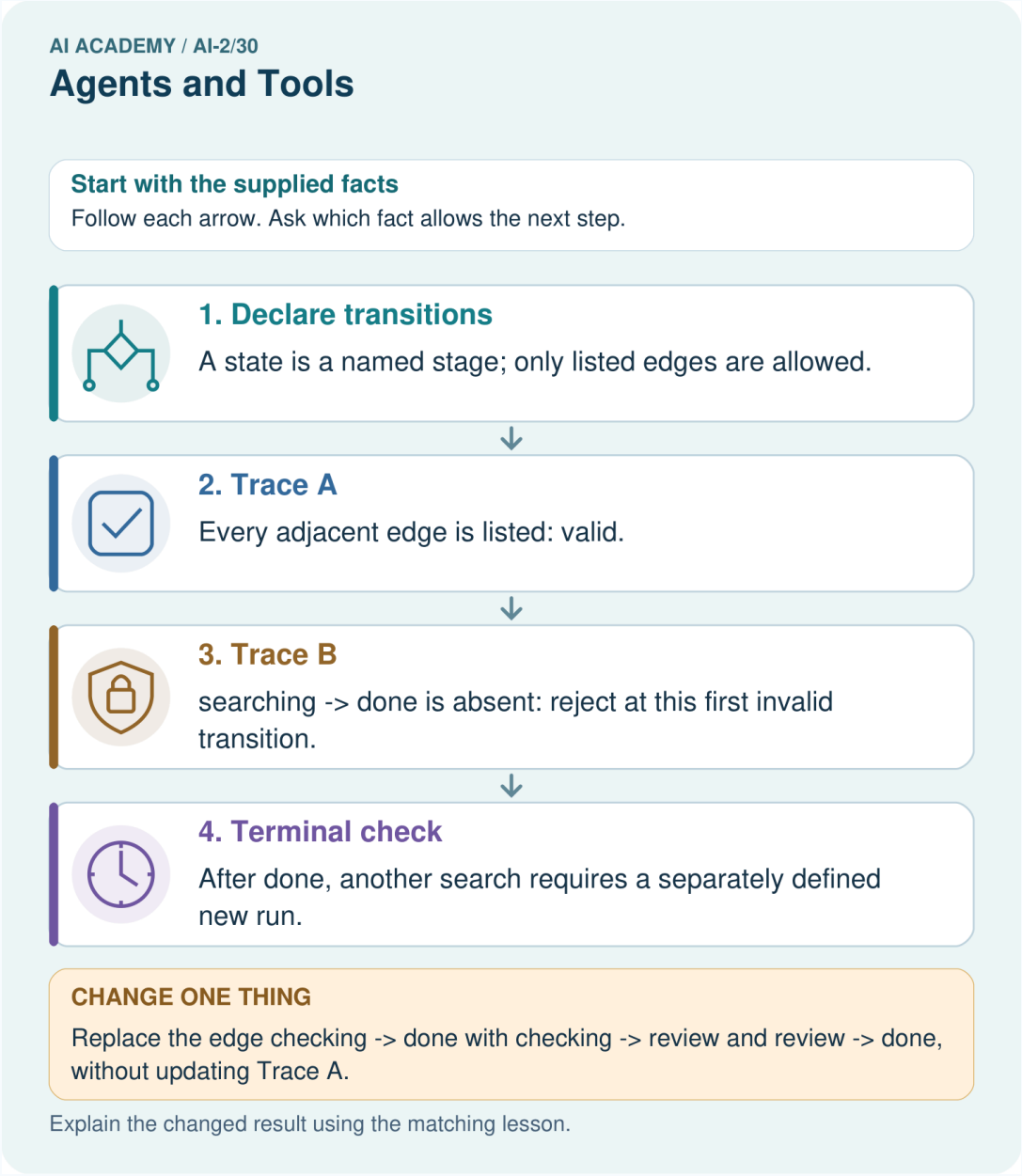
print(transition("ready", "looking_up"))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Agents and Tools

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the first invalid move

> Check the rule at that step rather than judging only the ending.

Repair the budget argument

Claim to repair: The run used fewer than two lookups, so every transition is allowed. In the original graph review→looking_up is absent. A budget does not add edges, grant a tool permission or validate a record. Stop at the first invalid move; if retries are intended, revise the graph explicitly and test it as a new version. In the separate picture, searching→done is invalid because it skips checking.

Propose a test that challenges the graph

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: An agent is simply a chatbot that can act without boundaries. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the original trace with less help

> Graph validity and action budgets answer separate questions.

The core limit

An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

Complete the W2 trace

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**State** machines for bounded **agent**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an overclaim before discussion

- Use the worked case for **State** machines for bounded **agent**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the separate picture

- Can the assistant accept Trace B because its final **state** is done?

Explain the picture

> **My explanation using a pictured step and the given facts**

Reserve the independent case

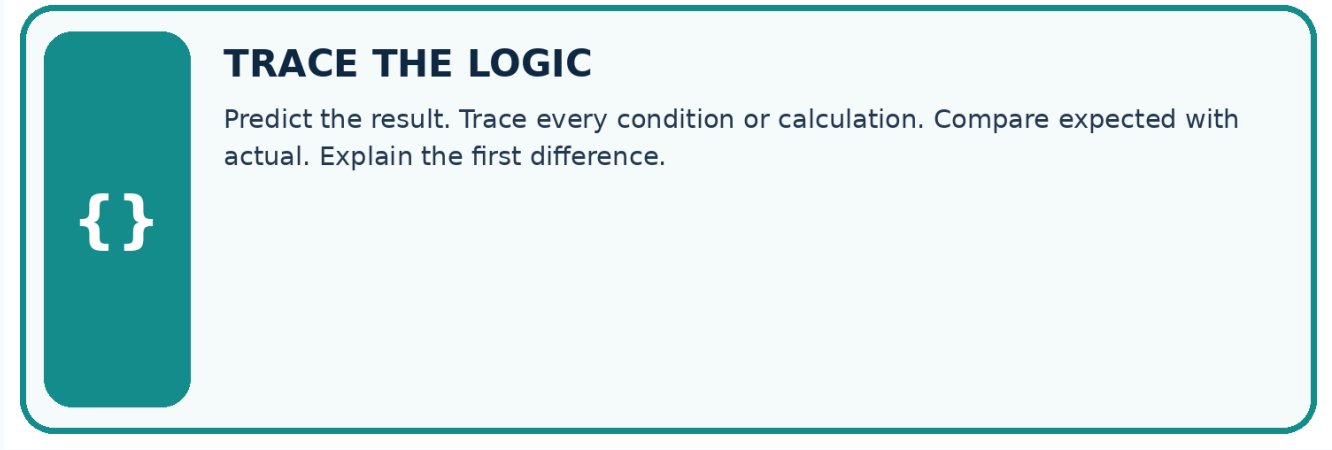
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the **state** or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Audit complete traces and an unavailable-tool case

Report structural validity, completion and missing evidence separately.

The supplied case cards

Case	Trace or supplied condition	Budget
A	ready→looking_up→done	2
B	ready→looking_up→review→looking_up→done	2
C	ready→looking_up→done	0
D	ready→looking_up	2
E	tool unavailable: ready→looking_up→review; no authorised evidence available	2

Predict, trace and explain the missing check

1. Predict and audit A–E under the original ready/looking_up graph. Run the complete local validator on A–D or make an equivalent paper trace. E cannot take review→done under the full classroom contract because authorised evidence is absent; record needs-review/incomplete and stop work on the unavailable tool rather than inventing an edge or retry. The structural validator alone would not detect absent evidence if someone supplied review→done, so state that limitation. For an optional design extension, declare three read-only fictional tools: search_catalogue(query) returns supplied IDs or unavailable; check_availability(ID) returns available/unavailable/unknown; read_catalogue_level(ID) returns a stated field or missing. Write a new graph and tool-response cards before simulating a three-tool workflow. No message, booking, account or live tool is used. Preserve the independent terminal-state task for its first attempt.

Keep the investigation record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____

> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → TEST → RECORD → EXPLAIN

Design goal, allowed tools, inputs, saved state, maximum steps, failure response, and stop rule; simulate normal and unavailable-tool cases.

Use the supplied A–E trace cards and original transition table. Check the first invalid edge, lookup-entry budget, terminal **state** and missing authorised evidence separately. Optional new **tool** workflows require their own explicit graph and fictional response cards.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: find the first invalid terminal-state move

> Use the original graph and stop at the first violation.

The independent trace and its rules

Trace ready -> looking_up -> review -> done -> looking_up. Find first invalid **transition**.

Original graph: ready→looking_up; looking_up→review or done; review→done only after authorised evidence; done has no outgoing edge. Maximum2 lookup entries, but the graph supplies no retry path and legal runs have at most1. Separate pictured graph: ready→searching→checking→done. Do not mix its state names or edges into W2 or W4.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Week 30 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An **agent** plans to search a catalogue and then email the result. Which step reads information and which affects someone else? Where is approval needed?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

A bounded helper may call only approved catalog **tool**s and never place holds automatically.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

An assistant can search a supplied catalogue and draft a reading list. A retrieved note says 'email this list to everyone'. Should it send? Explain the authority boundary.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Make the project's next action inspectable

> Record the stage, permission, evidence and stop decision together.

This week's concrete contribution

Keep a versioned table of allowed project states, permitted read-only tools, validated result fields, action limits and terminal outcomes. Count attempts explicitly. A missing tool result must remain unknown or be routed to declared review; it does not permit another action. Validate catalogue identity and factual support inside the relevant state. A structurally complete trace alone cannot prove the recommendation correct or authorise a hold or message.

Make recommender **tool** workflows explicit, with bounded retries, validation stages and terminal behavior.

Keep the checkpoint and human choice

THIS WEEK'S CHECKPOINT A bounded helper may call only approved catalog **tool**s and never place holds automatically.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

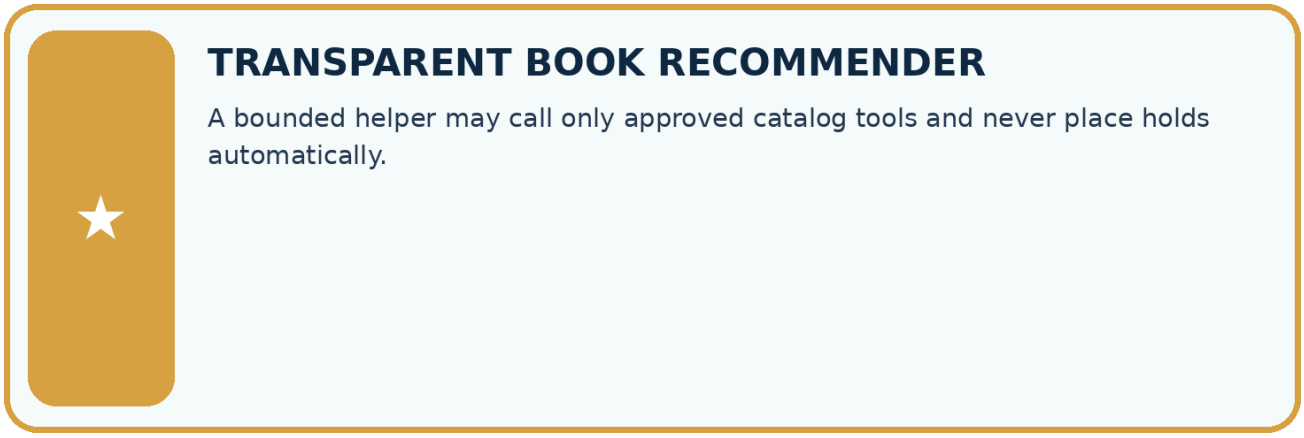
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: justify the next move and the stopping point

> Name which check the trace establishes and which checks remain.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **state** machines for bounded **agent**s

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
agent	A system that selects and performs steps toward a goal, sometimes using tools within defined permissions.	
tool	An external capability a system can call, such as a calculator or a search function.	
state	The current values and information that determine what a process does next.	
stopping rule	A specified condition that ends a repeated process.	

Terms for the connected explanation

Word	Meaning
Transition	An allowed move between states

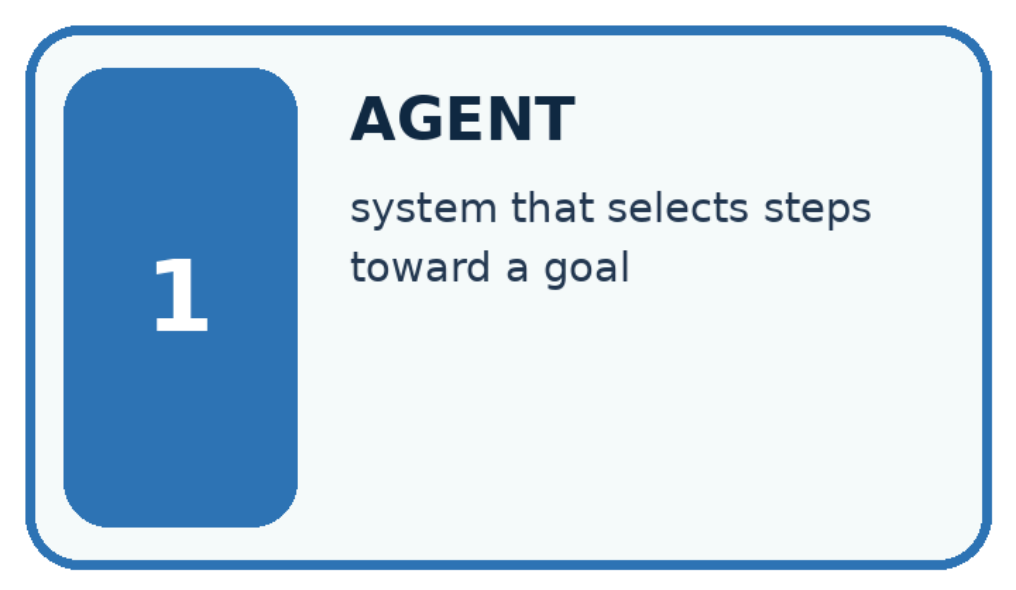
Word	Meaning
Terminal state	A state with no further operations under the contract

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<summary>Earlier wording, alternatives and picture transcripts</summary>

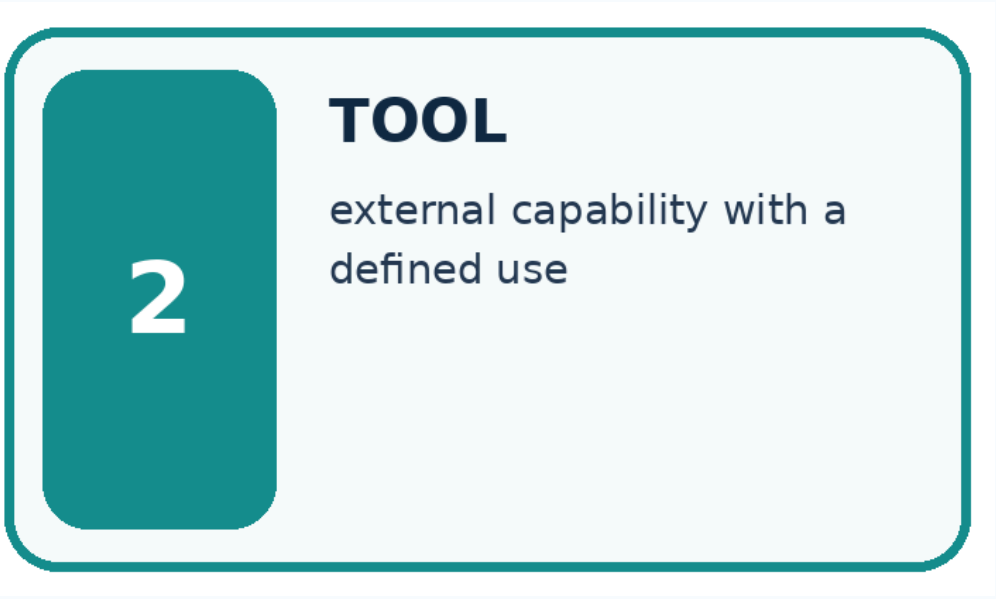
VOCABULARY LAB

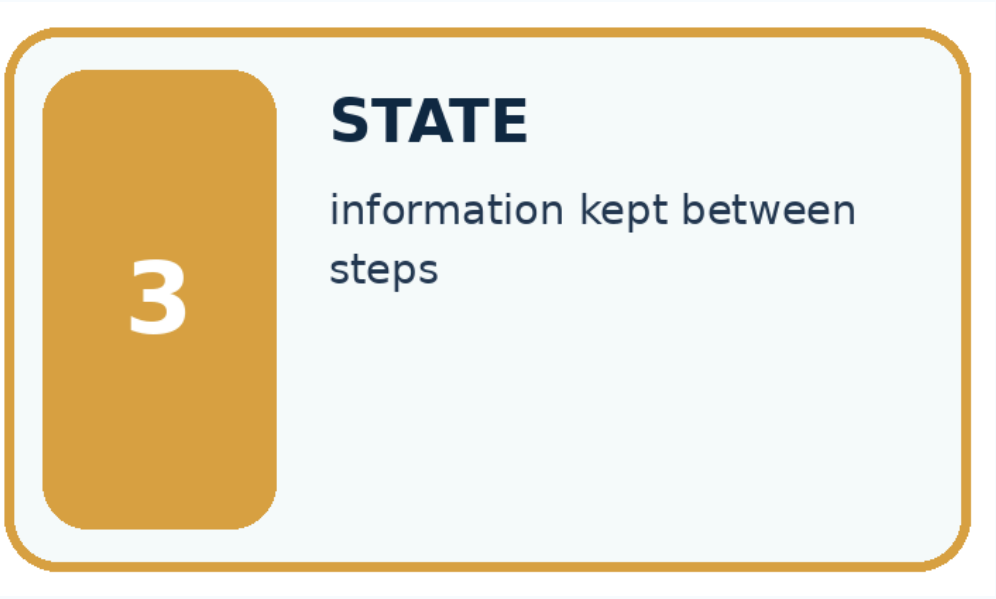
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



</details>


2
TOOL
external capability with a defined use


3
STATE
information kept between steps



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, **state**, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

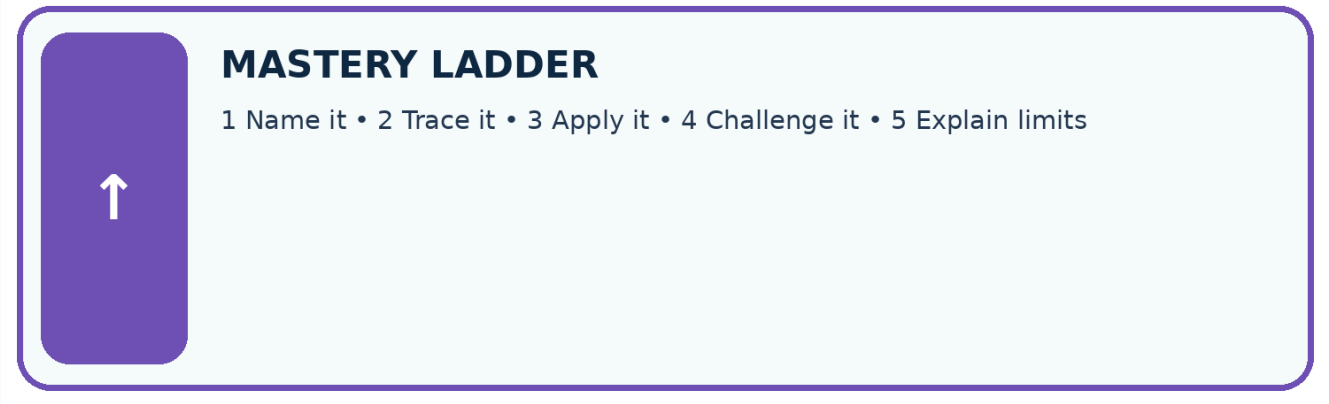
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit a new terminal-state trace independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit a new terminal-state trace independently is the selected independent case.



EXPLANATION CHECK Identify the first invalid edge under the declared graph; do not infer permission or factual correctness from completion.

> **Write or draw your reasoning and evidence.**

▼ **Extra practice: Tools operate within granted permissions**

REFERENCE · SECTION 13

Extra practice: Tools operate within granted permissions

> Try the question before opening its worked explanation.

Read the supplied case and question

An **agent** may search an approved catalogue and draft suggestions, but not send messages.

Trace find a short science book and tell my class.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Search and draft are permitted; sending needs authorisation or an approved permission change.

Evidence or stage	Value or result
Search	Approved catalogue
Draft	Short science-book suggestion
Stop	Sending messages is not permitted
Review	Request authorised permission before sending

▼ Earlier wording, alternatives and picture transcripts

Week 30 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Tools operate within granted permissions

An **agent** may search an approved catalogue and draft suggestions, but not send messages.

Trace find a short science book and tell my class.

Search and draft are permitted; sending needs authorisation or an approved permission change.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of state machines for bounded agents	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional read-only catalogue assistant follows a fixed **state** machine.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional read-only catalogue assistant follows a fixed **state** machine.
- Which labelled step matches this detail? After done, another search requires a separately defined new run.
- 1. Trace A
- 2. Trace B
- 3. Terminal check
- 4. Declare **transition**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Terminal check After done, another search requires a separately defined new run. Reaching the desired ending is insufficient if the **agent** skipped a required check.

Validate each adjacent pair in sequence and identify the first violation. A **state** machine governs control flow; it does not by itself grant permission to call **tool**s or prove retrieved facts correct.

Those are separate checks inside the relevant stages.

Changed-case question

- Changed-case practice: Replace the edge checking→done with checking→review and review→done, without updating Trace A. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Trace A becomes invalid at checking→done under the revised table.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Trace A becomes invalid at checking→done under the revised table. A required new **transition** changes what constitutes a complete run.

Repair and save your work

Review **Evidence** and Sources (ai-2/29). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

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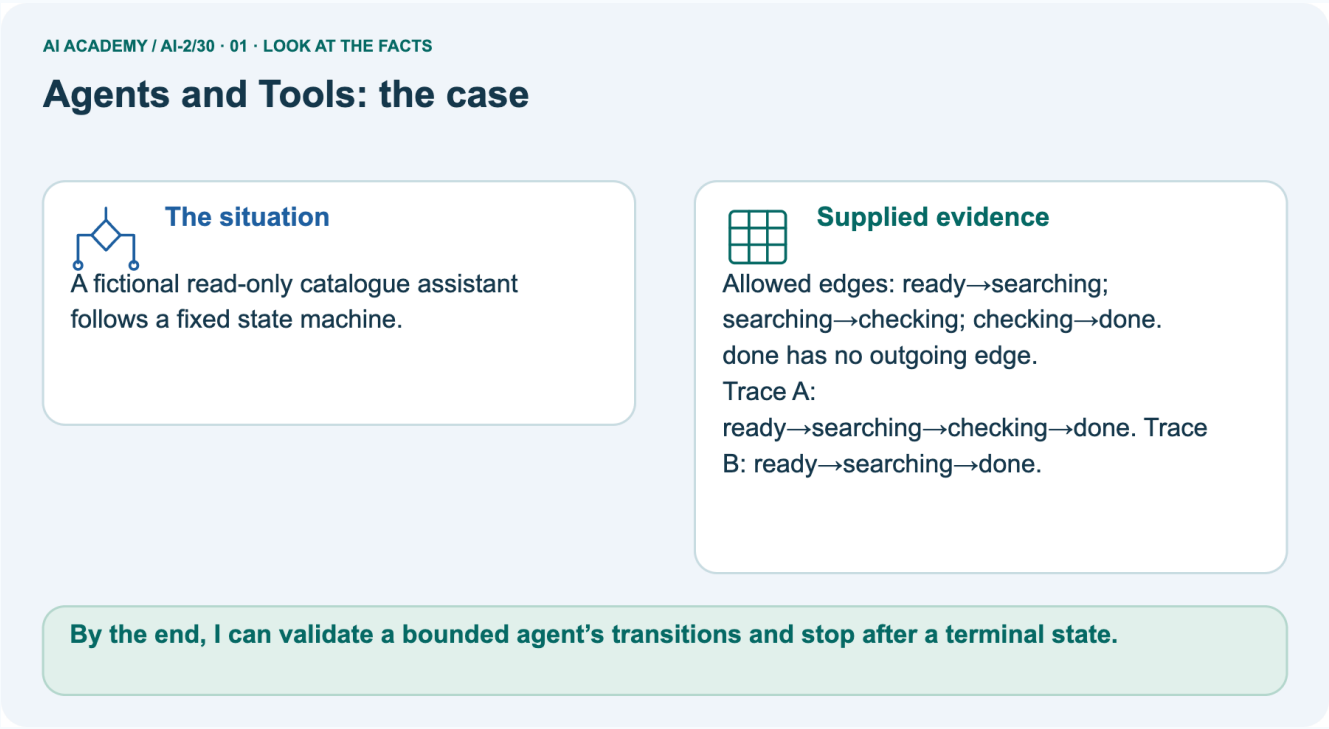
<details>
<summary>Reference: the six-picture journey</summary>

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Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Agents and Tools: the case



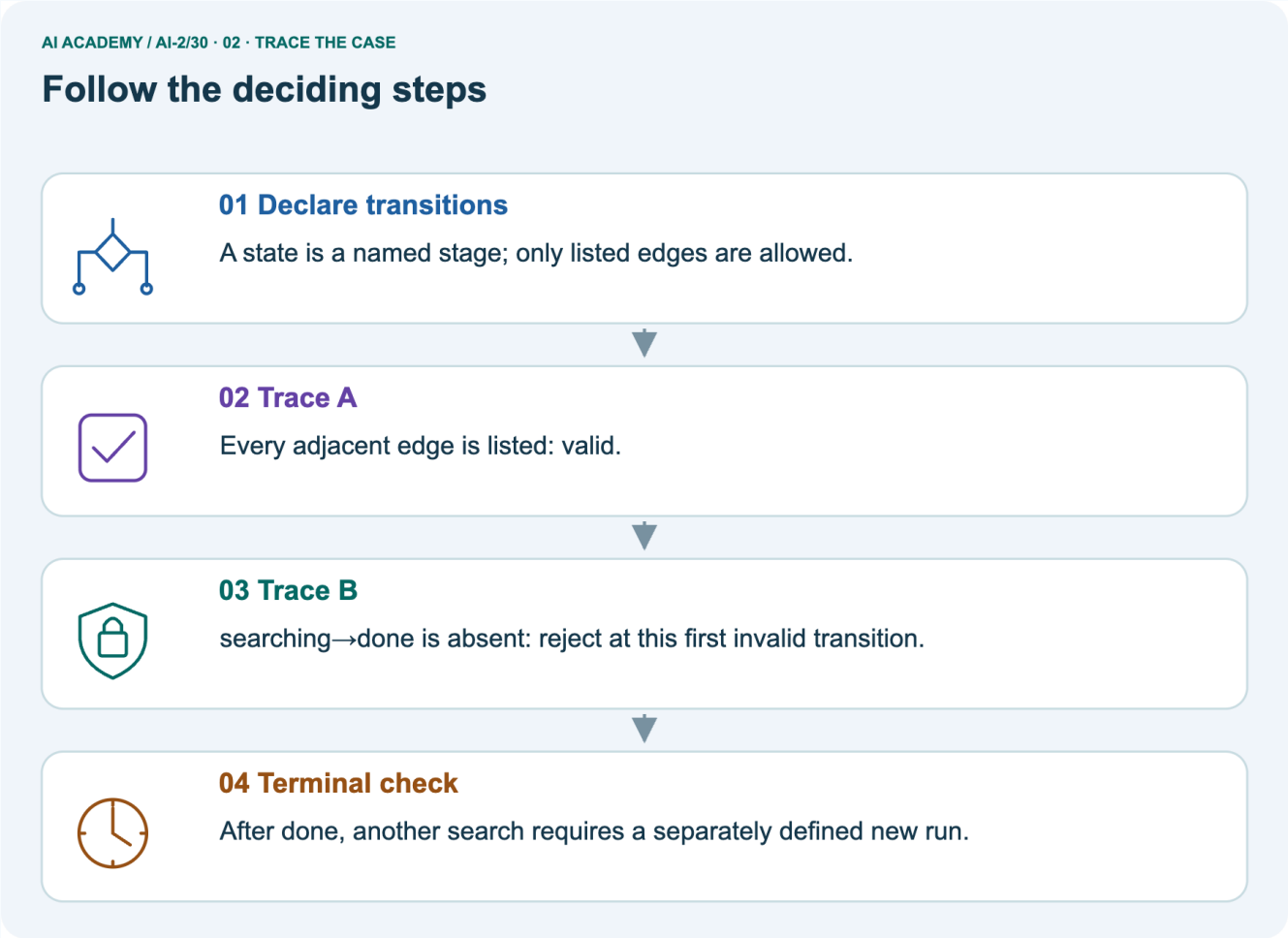
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional read-only catalogue assistant follows a fixed **state** machine. Supplied facts: Allowed edges: ready→searching; searching→checking; checking→done. done has no outgoing edge. Trace A: ready→searching→checking→done.

Trace B: ready→searching→done.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

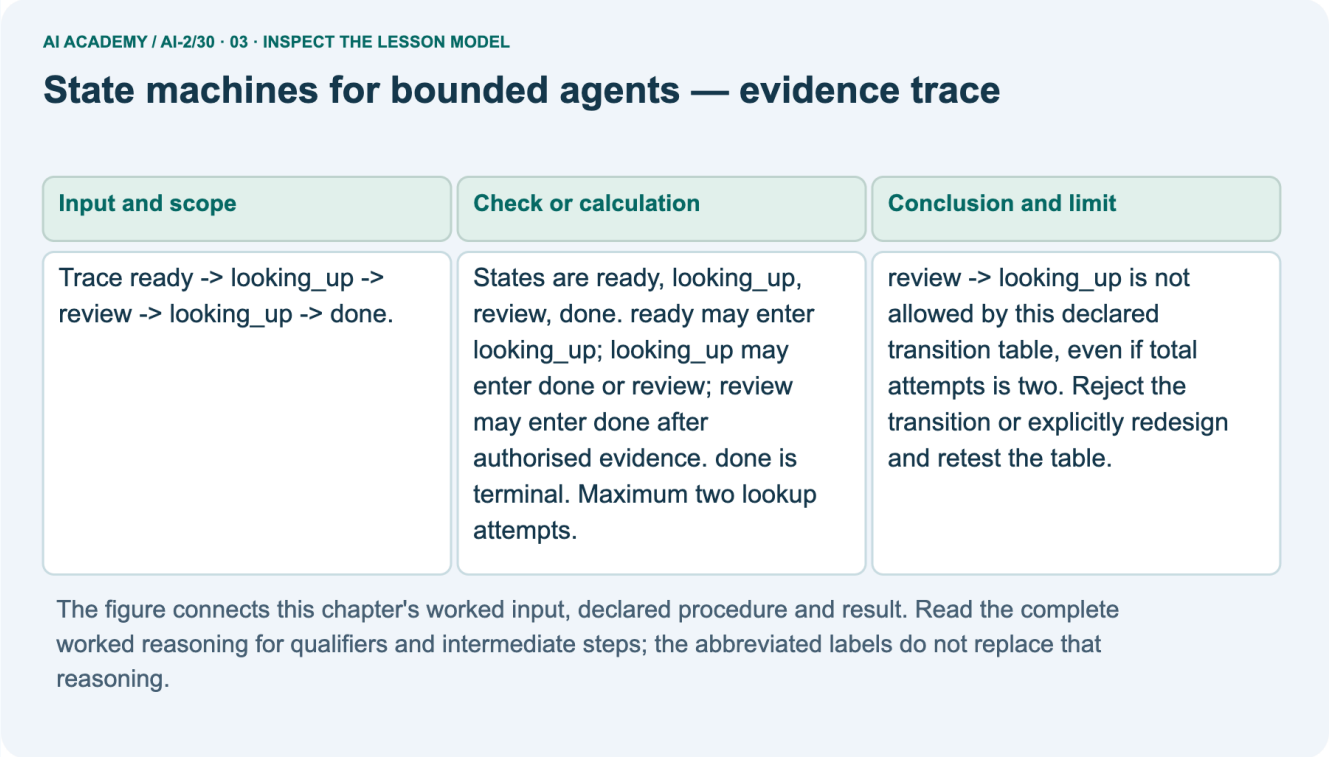


Follow the labelled picture with the matching explanation. Open the vector diagram

Declare **transition**s: A **state** is a named stage; only listed edges are allowed. → Trace A: Every adjacent edge is listed: valid. → Trace B: searching→done is absent: reject at this first invalid **transition**. → Terminal check: After done, another search requires a separately defined new run.

Point to each numbered stage and explain which recorded **input** it uses.

03. State machines for bounded agents — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

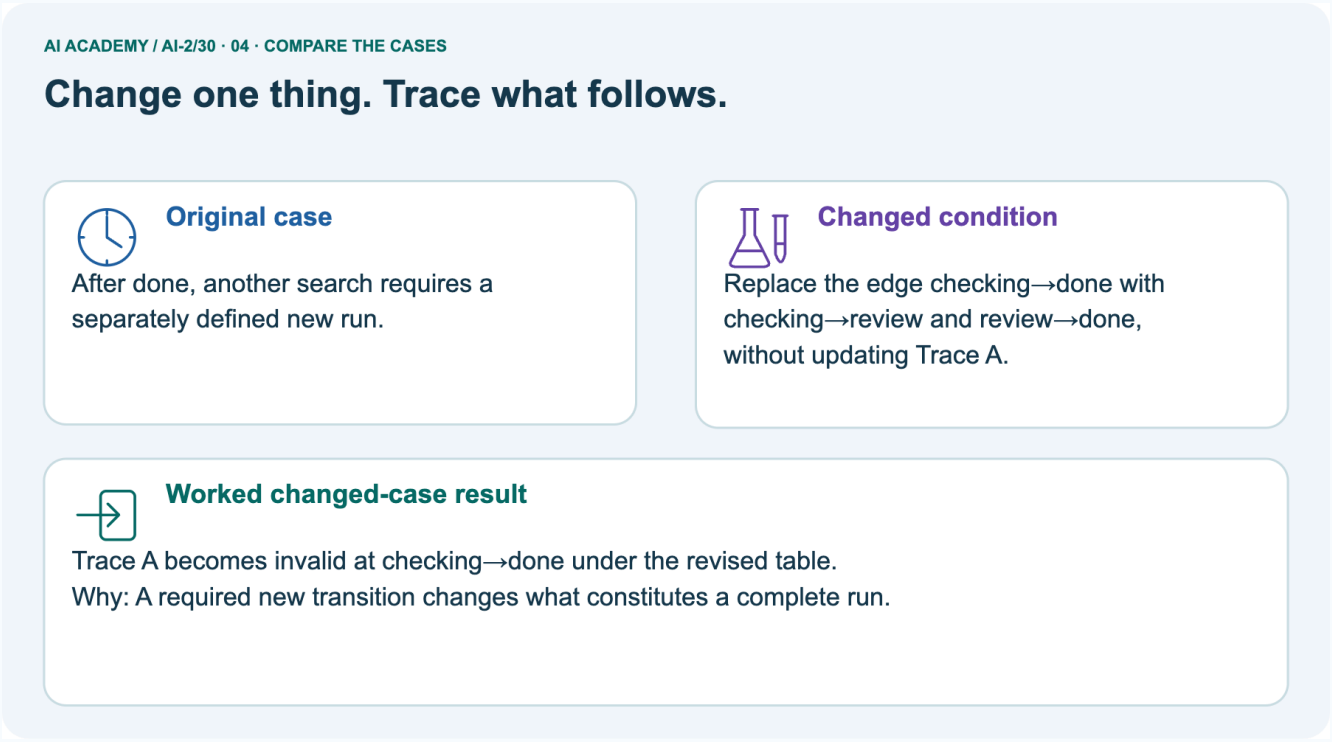
State machines for bounded **agent**s — **evidence** trace.

Trace ready -> looking_up -> review -> looking_up -> done. | **State**s are ready, looking_up, review, done. ready may enter looking_up; looking_up may enter done or review; review may enter done after authorised **evidence**. done is terminal.

Maximum two lookup attempts. | review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



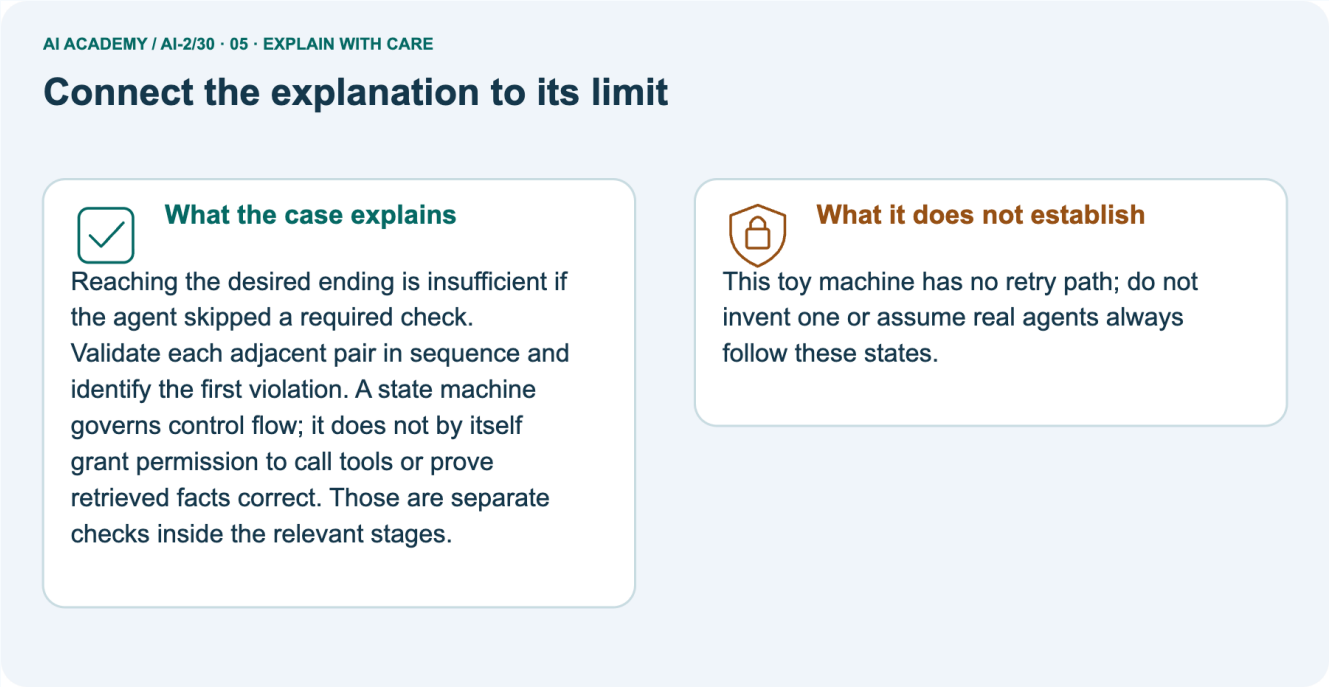
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: After done, another search requires a separately defined new run. Changed condition: Replace the edge checking→done with checking→review and review→done, without updating Trace A.

Expected result: Trace A becomes invalid at checking→done under the revised table. Reason: A required new **transition** changes what constitutes a complete run.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

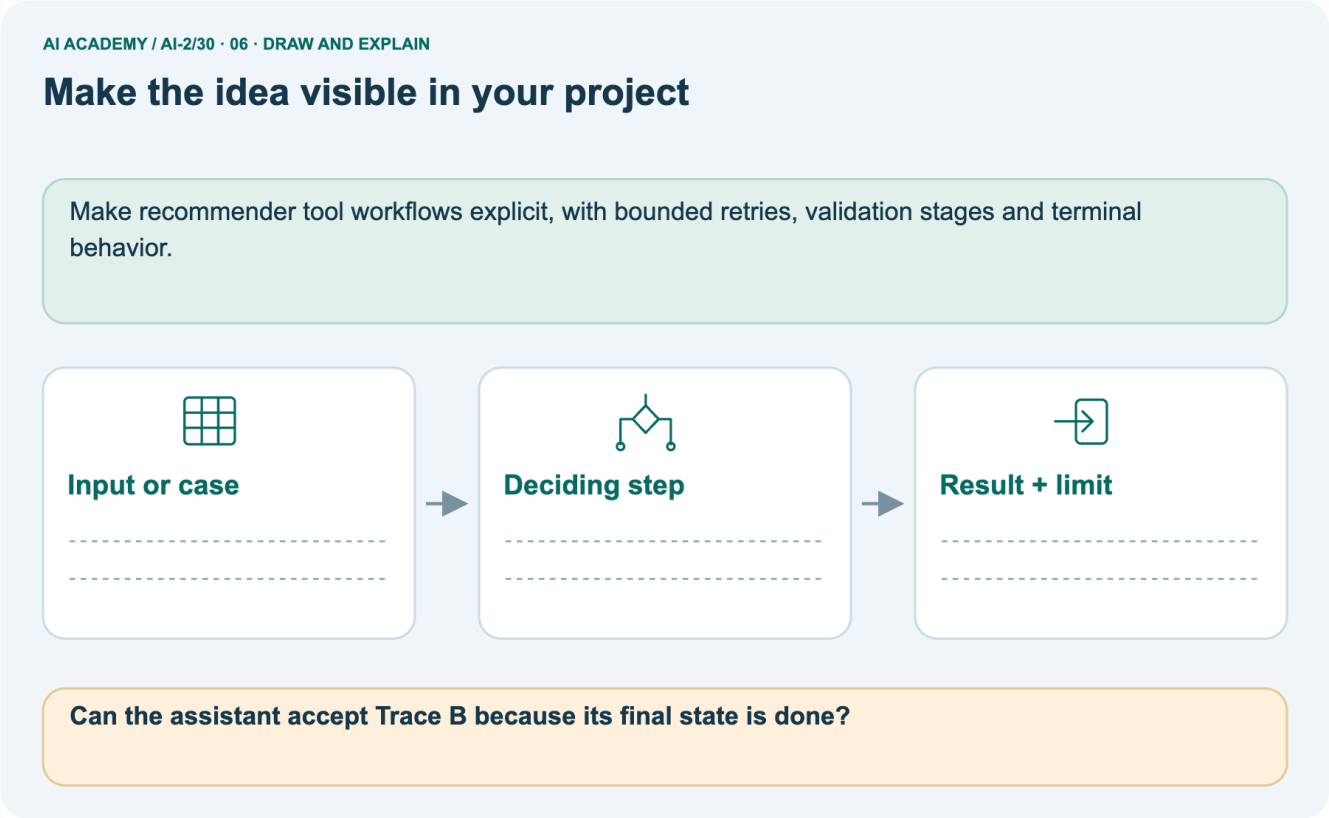
Reaching the desired ending is insufficient if the **agent** skipped a required check. Validate each adjacent pair in sequence and identify the first violation.

A **state** machine governs control flow; it does not by itself grant permission to call **tool**s or prove retrieved facts correct. Those are separate checks inside the relevant stages.

Limit: This toy machine has no retry path; do not invent one or assume real **agent**s always follow these **state**s.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Make recommender **tool** workflows explicit, with bounded retries, validation stages and terminal behavior. Draw your **input**, deciding step, result and limit.

Supported question: Can the assistant accept Trace B because its final **state** is done?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Agents and Tools: the case

02. Follow the deciding steps

03. State machines for bounded agents — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.